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BACHELOR OF SCIENCE IN MARINE ENGINEERING

**ENGINEERING DESIGN AND OPTIMIZATION OF PARAMETERIZED
MARINE STRUCTURAL COMPONENTS**

MR-PRO-26-XX

by

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for the award of the Degree of BACHELOR OF SCIENCE IN MARINE ENGINEERING
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DECLARATION

We hereby declare that this Project Report is our original work and has not been presented for a degree in any other university or for any other award.

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Dedication

We dedicate this Project Report to our parents, guardians, lecturers, and mentors whose encouragement, sacrifice, and guidance have supported our academic journey.

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We would like to express our sincere gratitude to all those who contributed to the completion of this Project Report. We extend our deepest appreciation to our supervisor for invaluable guidance and constructive feedback.

Abstract

This project investigated the constrained mass optimization of marine structural components using parameterized geometry and first-order structural models. The primary case was an Aluminium 6061-T6 equipment bracket carrying a 50 kg auxiliary unit under a 1471.5 N design load. A 30-point Latin-hypercube study and two bounded Nelder–Mead searches were completed. The baseline analytical mass was 0.4181 kg; the lightest screened candidate had a mass of 0.1126 kg, factor of safety 2.60, and deflection 0.361 mm, giving a 73.1% analytical mass reduction. The method was extended to an S275 lifting padeye and a fixed-envelope aluminium stabilizer fin through engineering definitions, analytical models, CAD generators, and validation packages. Fusion stress results were not recorded in the available validation tables; consequently, reported stress and deflection values are screening predictions requiring mesh-converged FEA and physical validation before fabrication.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

FEA	Finite Element Analysis
CAD	Computer-Aided Design
FOS	Factor of Safety
API	Application Programming Interface
IPC	Inter-Process Communication
LHS	Latin-Hypercube Sampling

Symbols

m	Mass (kg)
σ	Stress (MPa)
g	Acceleration due to gravity (m/s ²)
δ	Deflection (mm)

Chapter 1

Introduction

Marine vessels contain numerous brackets, foundations, lifting fittings, and control-surface structures. These components must transmit demanding loads without yielding or excessive deformation while contributing as little unnecessary mass as practicable. Excess mass increases material consumption and vessel displacement, whereas excessive material removal reduces stiffness and fatigue resistance. Marine structural design is therefore a constrained engineering problem.

1.1 Problem Statement

Conventional component design uses hand calculations, conservative initial sizing, CAD modelling, and repeated checks. Only a limited number of dimensional combinations can normally be examined manually. A systematic method was required to explore plate dimensions, stiffener sizes, and fillet radii while enforcing strength and deflection limits.

1.2 Objectives

The main objective was to develop and assess a parameterized engineering method for lower-mass marine structural components. The specific objectives were to:

1. define materials, loads, boundary conditions, variables, and acceptance criteria;
2. formulate analytical models for mass, stress, factor of safety, and deflection;
3. explore the bracket design space by Latin-hypercube sampling and bounded Nelder–Mead optimization;
4. generate parameterized geometry and validation candidates in Fusion 360;
5. extend the method to a lifting padeye and fixed-envelope stabilizer fin; and
6. identify the detailed FEA and physical validation required before practical adoption.

1.3 Scope and Significance

Completed work covered linear-elastic analytical screening, continuous dimensional variables, CAD generation, and validation preparation. Fatigue, buckling, weld-detail stress, corrosion, nonlinear contact, vibration, impact, CFD, fabrication, and proof testing were outside the completed scope. The engineering value lies in traceable comparison of alternatives, not in replacing detailed analysis or professional judgement.

Chapter 2

Literature Review

The pursuit of optimal structural design has been a cornerstone of mechanical and marine engineering for decades, representing an ongoing evolution from empirical craftsmanship to highly formalized computational science. Historically, this design process relied heavily on manual iteration—a workflow where engineers intuitively adjusted design parameters based on rules of thumb, basic analytical calculations, and eventually, manual Finite Element Analysis (FEA) results until a satisfactory, albeit not strictly optimal, solution was found [1]. However, the increasing complexity of modern structural systems, coupled with the pressing global need for material efficiency, drastic cost reduction, and performance maximization under rigorous environmental constraints, have necessitated the adoption of automated, algorithmic approaches to design [2]. The transition towards such methods marks a departure from finding merely "feasible" designs to mathematically securing "optimal" configurations.

This chapter comprehensively reviews the theoretical background, historical context, and state-of-the-art developments across several intersecting domains essential to the Nelder-Mead Marine Optimizer (NEMO) project. First, it examines the evolution and automation of parametric Computer-Aided Design (CAD) and FEA, exploring how Application Programming Interfaces (APIs) have revolutionized data exchange. Second, it investigates the application and necessity of gradient-free optimization algorithms within structural engineering, detailing why traditional derivative-based methods falter in noisy design spaces. Third, it provides a deep dive into the history, mathematics, and convergence properties of the Nelder-Mead simplex method, contrasting it with population-based metaheuristics like Genetic Algorithms (GAs). Fourth, the role of surrogate modeling in bridging the computational divide between high-fidelity FEA and optimization is discussed. Finally, it addresses the specific challenges, historical trajectories, and contemporary methodologies characterizing structural optimization within marine and maritime applications. By synthesizing these diverse areas, this literature review identifies the current limitations of conventional workflows and explicitly defines the critical research gap that the NEMO project aims to bridge: the democratization of component-level

structural optimization in the maritime industry.

2.1 Automation in Parametric CAD and FEA

The integration and automation of parametric CAD and FEA represent a paradigm shift from traditional, serial design workflows to closed-loop, algorithm-driven design spaces. In early computer-aided design, geometric models were essentially static digital drawings; modifying a single dimension required painstakingly redrawing connected elements. The advent of parametric modeling revolutionized this by allowing engineers to define component geometries through associative variable parameters (such as lengths, thicknesses, radii, and angular constraints) rather than rigid, static Cartesian coordinates. When this parametric capability is tightly integrated with an automated FEA pipeline, a change in a single parameter can trigger a cascading, automatic update of the geometry, followed by mesh regeneration, and subsequent structural analysis [3].

Recent literature highlights the immense value of this automation in drastically reducing the “time-to-insight” for design evaluations. Manual intervention in the geometry cleanup, model idealization, and meshing phases has historically been the most significant and time-consuming bottleneck in the engineering simulation pipeline. According to Song et al. [4], the development of robust Application Programming Interfaces (APIs) for commercial CAD software has empowered engineers to bypass graphical user interface (GUI) driven manual interactions entirely. By scripting these processes using languages such as Python, C#, or C++, a “closed-loop” environment is established where CAD geometry generation, FEA evaluation, and algorithmic decision-making communicate seamlessly without human oversight [5].

However, achieving true interoperability remains a persistent and significant technical challenge. Transferring geometric data between disparate, standalone CAD and FEA software packages typically relies on neutral file formats like STEP (Standard for the Exchange of Product model data) or IGES. This translation often results in subtle information loss, the generation of sliver faces, or the breaking of associative boundary condition links—phenomena that can abruptly derail automated optimization routines [3]. Furthermore, maintaining a “fit-for-purpose” analytical model—where parameterized, idealized geometry is reliably updated without causing mesh element inversion, aspect ratio failures, or macroscopic topological violations—is technically demanding. If a parameter change causes a hole to intersect an edge, the resulting topological shift can cause automated meshing scripts to crash.

To circumvent these issues, modern computational design approaches favor utilizing unified software platforms or highly integrated, native APIs. For example, the Autodesk Fusion 360 Add-Ins API provides a unified environment where both the solid modeling kernel and the simulation solver reside within the same data structure, ensuring that parametric relationships, constraints, and structural boundary conditions remain entirely

invariant throughout thousands of iteration cycles [5]. The NEMO project deliberately leverages this integrated architecture to systematically avoid the pitfalls of cross-platform data exchange. By utilizing an embedded Python environment to orchestrate the geometric generation and structural evaluations autonomously, the project ensures a highly robust, fault-tolerant pipeline capable of sustaining continuous algorithmic iteration.

2.2 Gradient-Free Optimization Algorithms in Structural Engineering

Optimization within the realm of structural engineering is inherently mathematically challenging due to the specific nature of the objective functions and constraint boundaries involved. In almost all practical applications, these functions do not exist as neat algebraic equations; rather, they represent the discrete numerical outputs of complex, computationally expensive, “black-box” FEA simulations. In such structural scenarios, the relationship between input design variables (e.g., the thickness of a web, the radius of a stress-relieving fillet, the spacing of stiffeners) and the resulting structural responses (e.g., maximum localized von Mises stress, peak structural deflection, fundamental frequency) is highly non-linear, frequently discontinuous, and critically lacks explicit analytical derivatives [6].

Gradient-based optimization methods (such as Sequential Quadratic Programming or Interior Point methods), while exceptionally fast and mathematically rigorous, fundamentally require smooth, continuous, and differentiable objective spaces to compute analytical sensitivities (Jacobians and Hessians) [7]. When exact gradients are completely unavailable, engineers attempt to approximate them numerically using finite difference methods. However, in FEA-driven problems, this numerical approximation is highly susceptible to artificial “noise” generated by meshing artifacts. As the geometry slightly changes, the discrete mesh topology shifts, causing microscopic, artificial spikes and dips in the resulting stress calculations. This high-frequency noise entirely confounds gradient-based solvers, causing them to calculate wildly inaccurate descent directions, fail to converge, or become hopelessly trapped in poor local optima [8].

Consequently, gradient-free (also known as derivative-free or direct search) optimization algorithms have transitioned from being viewed as mathematically unrefined to becoming the absolute standard practice for complex, simulation-based structural problems [9]. As Wright [8] chronicles in her seminal review of direct search methods, the realization that engineering functions are inherently noisy and non-differentiable led to a massive resurgence in the popularity and respectability of algorithms that rely solely on evaluating the objective function itself, bypassing the need for gradient estimation entirely.

2.3 The Nelder-Mead Simplex Method: History and Mathematical Convergence

Among the vast array of derivative-free optimization techniques, the Nelder-Mead (NM) simplex algorithm stands as one of the most historically significant, resilient, and widely utilized direct-search methods. The method was originally published in 1965 by John A. Nelder and Roger Mead [10]. It was developed as a direct, adaptive improvement upon an earlier “simplex” search strategy proposed by William Spendley, George Hext, and Frank Himsworth in 1962 [11]. While the original Spendley et al. method utilized a rigid, regular simplex that maintained a constant geometric size and shape as it moved through the design space, Nelder and Mead introduced the revolutionary concept of allowing the simplex to dynamically adapt its shape—stretching to accelerate down steep valleys, contracting to navigate narrow corridors, or shrinking to finely pinpoint a local minimum [12]. It is critical to clarify that the Nelder-Mead method is entirely unrelated to George Dantzig’s famous “Simplex Method” used for solving constrained linear programming problems; Nelder-Mead is explicitly designed for unconstrained (or penalty-constrained) non-linear optimization [9].

The algorithm navigates the mathematical design space using a simplex—a geometric figure consisting of $n + 1$ vertices in an n -dimensional space (e.g., a triangle in 2D space, a tetrahedron in 3D space). The algorithm begins by evaluating the objective function at each of these initial vertices. It then iteratively attempts to replace the vertex yielding the worst (highest) function value with a new, better point. This is achieved through a sequence of specific geometric transformations:

1. **Reflection:** The worst point is reflected across the centroid of the remaining points, projecting it into a potentially better region of the design space.
2. **Expansion:** If the reflected point is highly successful (better than the current best point), the algorithm expands the simplex further in that direction, accelerating the search.
3. **Contraction:** If the reflection fails to yield a better point, the algorithm assumes it has overstepped a valley and contracts the simplex inwards, probing closer to the centroid.
4. **Shrinkage:** If all else fails, the entire simplex shrinks uniformly towards the best known vertex, effectively reducing the step size to finely resolve the local topology.

The primary advantage of the Nelder-Mead algorithm is its absolute indifference to gradient information and its remarkable robustness in the face of the numerical noise typical of FEA outputs. Kaveh and Ghazaan [13] demonstrated the profound efficacy of

the Nelder-Mead algorithm in continuous structural design, noting its inherent simplicity, robustness against discontinuities, and ease of algorithmic implementation.

However, the mathematical convergence properties of the Nelder-Mead method have been a subject of intense academic scrutiny. Despite its immense practical success, it lacks the strong theoretical convergence guarantees found in gradient-based methods [14]. Lagarias et al. [15] conducted a rigorous analysis of the method’s convergence properties, proving convergence for strictly convex functions in 1D and 2D spaces, but noting that in higher dimensions, the algorithm can, under highly specific adversarial conditions, stall and converge to a non-stationary point that is not a true local minimum. Furthermore, as a strictly local search method, Nelder-Mead is fundamentally vulnerable to becoming trapped in local minima if the initial starting simplex is situated in a poor region of a highly multimodal landscape. Despite these theoretical limitations, its empirical performance on smooth, low-dimensional engineering problems remains unparalleled, making it highly attractive for specific classes of structural refinement [12].

2.4 Genetic Algorithms vs. Nelder-Mead in Structural Optimization

In the contemporary landscape of structural optimization, the comparison between population-based metaheuristics, most notably Genetic Algorithms (GAs), and local direct-search methods like Nelder-Mead is characterized by a fundamental trade-off between global exploration and local exploitation [16].

Genetic Algorithms, such as the highly influential Non-dominated Sorting Genetic Algorithm II (NSGA-II) [17], are inspired by the principles of Darwinian natural selection. They operate by maintaining a large “population” of candidate designs. Through iterative cycles of evaluating fitness, selecting the best performers, and applying biologically inspired operators like “crossover” (recombination of traits) and “mutation” (random perturbations), GAs comprehensively explore vast, complex, and highly multimodal design spaces. They are exceptionally robust, easily handle discrete variables (like standardized plate thicknesses), and are highly effective at avoiding local optima to locate the global optimum region.

However, the defining limitation of Genetic Algorithms in the context of FEA-driven structural optimization is their abysmal sample efficiency. Because they rely on statistical populations and randomized exploration, GAs typically require hundreds, if not thousands, or tens of thousands of individual function evaluations to achieve convergence [6]. When each evaluation involves a computationally expensive 3D meshing and finite element solve—taking minutes or hours—the computational cost of deploying a GA becomes entirely prohibitive for routine engineering tasks without access to massive high-performance computing (HPC) clusters. Furthermore, GAs are notoriously slow

at "fine-tuning"; once they have identified the general valley containing the global optimum, their randomized nature makes pinpointing the exact mathematical minimum highly inefficient [16].

Conversely, the Nelder-Mead method acts as a highly aggressive, deterministic local search. It is engineered for rapid exploitation. It requires significantly fewer function evaluations to "slide down" a valley and locate a precise minimum, making it vastly more sample-efficient than any population-based method [13]. The cost of this speed is the loss of global awareness; Nelder-Mead will reliably find the bottom of the valley it starts in, but it cannot know if a deeper valley exists elsewhere in the design space.

In advanced aerospace and automotive research, the prevailing state-of-the-art favors utilizing hybrid algorithms that combine the strengths of both [6]. A GA is deployed initially to explore the space and identify promising "basins of attraction," after which the Nelder-Mead algorithm is seeded in these basins to rapidly exploit and converge on the exact optimum. However, for the NEMO project—which specifically targets localized, secondary marine components (such as brackets and stiffeners) rather than entire hull forms—the design spaces are typically low-dimensional (e.g., 2 to 6 continuous variables) and generally unimodal (or possess local optima that are sufficiently "good" for engineering purposes). In this highly specific context, where computational economy on a standard engineering workstation is paramount, the deterministic speed and high sample efficiency of the Nelder-Mead method present a definitively superior primary optimization strategy over computationally exorbitant GAs.

2.5 Surrogate Modeling in FEA: Bridging the Computational Divide

As the ambition of structural optimization has grown, so too has the computational burden of high-fidelity FEA. When a single simulation of a complex assembly or non-linear crash analysis takes days to compute, even sample-efficient algorithms like Nelder-Mead become impractical [18]. To bridge this computational divide, researchers and engineers heavily employ Surrogate Modeling—also known as metamodeling or response surface construction [19].

Surrogate models act as mathematically cheap, "black-box" analytical approximations of the expensive FEA solver. The standard workflow involves defining a Design of Experiments (DOE)—such as a Latin Hypercube Sampling [20]—to intelligently select a limited number of training points across the multidimensional design space. The expensive FEA solver is run only at these specific points. Using the resulting input-output data pairs, regression or interpolation techniques are utilized to construct an analytical mathematical surface that mimics the FEA behavior [21].

Two of the most prominent surrogate techniques in structural engineering are Re-

sponse Surface Methodology (RSM) and Kriging (Gaussian Process Regression). Traditional RSM typically utilizes low-order (often quadratic) polynomials to formulate a global approximation of the design space using least-squares regression. It is computationally trivial to construct and evaluate, making it excellent for identifying broad, global trends in relatively smooth, low-dimensional spaces [19]. However, RSM generally fails to accurately capture highly complex, multi-modal, or highly non-linear structural behaviors.

Kriging, originally developed in geostatistics, has emerged as the premier surrogate modeling technique for advanced structural optimization [18]. Kriging provides a highly flexible, spatial statistics-based approach that combines a global polynomial model with a localized deviation function (a Gaussian process). This allows Kriging to act as an exact interpolator that precisely passes through the sampled FEA data points while robustly estimating the uncertainty and non-linear behavior in the unexplored regions between them [21].

When integrated into an optimization loop, surrogate models offer profound benefits. First, they dramatically accelerate the optimization process, as the algorithm (whether GA or NM) interrogates the instantaneous mathematical surrogate rather than invoking the slow FEA solver. Second, regression-based surrogates inherently "smooth out" the high-frequency numerical noise associated with FEA meshing artifacts, providing a clean, differentiable surface that allows even gradient-based optimizers to function effectively [18]. Finally, the explicit mathematical nature of the surrogate allows engineers to rapidly perform sensitivity analyses, instantaneously visualizing how structural performance varies with design changes.

While surrogate modeling is absolutely essential for large-scale, system-level maritime optimization (e.g., full-ship dynamic load analysis), the NEMO project explicitly focuses on component-level optimization where individual linear-static FEA solves can be executed in a matter of seconds. Consequently, NEMO opts for direct algorithmic evaluation, eliminating the complexity and potential approximation errors of constructing surrogate models, while acknowledging their critical role in broader maritime optimization contexts.

2.6 Structural Optimization in Marine and Maritime Applications

The maritime industry presents uniquely severe structural challenges characterized by profoundly harsh operational environments, highly dynamic and unpredictable loading conditions (wave slamming, torsional hull flexing, cargo inertia), and an incredibly strict, conservative regulatory framework enforced by classification societies [22]. The historical evolution of marine structural design reflects a slow, cautious transition from reliance on empirical, craftsmanship-based shipwright rules to the adoption of advanced, computa-

tionally intensive, multi-objective design frameworks [23].

Traditionally, marine structures—from the massive primary longitudinal hull girders down to the smallest secondary auxiliary outfitting brackets—have been systematically over-designed. Driven by the overriding priority of absolute reliability at sea, engineers historically employed massive safety margins to compensate for the inherent uncertainties in dynamic oceanic loads and material fatigue degradation over a 25-year lifespan [2]. However, modern economic and environmental realities have radically shifted the industry’s priorities. The primary objective of contemporary marine structural optimization is to relentlessly minimize steel weight, known as lightweight tonnage (LWT). Reducing LWT directly correlates to massive reductions in initial material procurement costs, an increase in profitable cargo deadweight capacity, lowered vessel displacement, drastically improved hydrodynamic fuel efficiency, and critical reductions in greenhouse gas (GHG) emissions [22]. The implementation of the International Maritime Organization’s (IMO) Energy Efficiency Design Index (EEDI) has transformed structural lightweighting from an economic desire into a strict regulatory mandate.

Cui and Zhang [22] highlight that Simulation-Based Design Optimization (SBDO) is increasingly utilized across the marine sector to systematically strip away this historical excess mass. By establishing formal mathematical constraints based on yield strength criteria, buckling resistance, and high-cycle fatigue life under extreme load cases, optimization algorithms can safely iteratively remove unnecessary material from non-critical load paths without compromising the vessel’s structural integrity or violating classification society rules.

Current marine optimization literature predominantly categorizes efforts into three methodologies. First, Scantling Optimization focuses on the iterative adjustment of plate thicknesses and standard stiffener profiles (e.g., bulb flats) to meet classification rules with minimum mass [23]. Second, Shape Optimization adjusts the geometric boundaries of components, such as optimizing the curvature of a bracket edge to reduce stress concentrations. Third, Topology Optimization radically redistributes material within a given design domain to create highly organic, weight-efficient internal structures, although these often face severe manufacturability challenges in traditional shipyards.

Despite the proliferation of sophisticated optimization techniques at the global ship-design level—such as holistic hull form optimization using Computational Fluid Dynamics (CFD) to minimize wave-making resistance, or midship section scantling optimization to maximize longitudinal bending strength—the optimization of secondary structural components and outfitting remains stubbornly manual and archaic. These localized components, such as piping supports, equipment foundations, and mounting brackets, exist in the hundreds or thousands aboard a typical commercial vessel. While the theoretical mass saving on a single bracket may appear entirely negligible (perhaps a few kilograms), the aggregate weight reduction across a ship’s entire massive outfitting inventory represents a highly substantial and entirely unexploited opportunity for efficiency gains [22]. Fur-

thermore, these localized components, governed by clear, isolated boundary conditions, are exceptionally amenable to parameterization and rapid automated structural analysis.

2.7 Research Gap and Motivation

A critical and exhaustive review of the existing literature reveals a profound and distinct gap between the theoretical, highly developed capabilities of automated structural optimization and its practical, day-to-day application to secondary marine structures using accessible engineering tools. While the advanced mathematical theory of derivative-free optimization [9, 13] and the architectural frameworks for parametric CAD/FEA API integration [3, 4] are incredibly well-established in academic literature, these powerful domains are remarkably rarely synthesized into lightweight, deployable, and user-friendly pipelines suitable for everyday shipyard engineering components.

The prevailing state-of-the-art in maritime research predominantly focuses either on massive, global ship structures using highly specialized, exceptionally expensive, and notoriously difficult-to-learn enterprise maritime software suites, or alternatively, on the purely abstract mathematical benchmarking of novel algorithms without a tangible physical CAD/FEA loop attached [14]. Consequently, a heavy reliance on manual, intuition-driven trial-and-error iteration violently persists in localized marine component design. The establishment of an automated optimization pipeline is widely perceived by practicing shipyard engineers as too programmatically complex, computationally exorbitant, or software-prohibitive for routine use on "minor" components.

NEMO addresses this gap through analytical screening, bounded Nelder–Mead search, automated Fusion geometry generation, and documented manual FEA validation. Its file-based handshake preserves traceable requests and responses while separating the optimizer from Fusion’s embedded environment. The study evaluates reliable partial automation while retaining engineering review where the current API does not support dependable solver control or result extraction.

Chapter 3

Engineering Methodology

3.1 Bracket Definition

The bracket represented a foundation for a 50 kg auxiliary machine. Aluminium 6061-T6 was assigned density 2700 kg/m³, yield strength 276 MPa, elastic modulus 68.9 GPa, and Poisson ratio 0.33. The adopted load was

$$F_d = mg\gamma_d = 50(9.81)(3) = 1471.5 \text{ N}, \quad (3.1)$$

where the dynamic factor of 3 is an engineering envelope rather than a modal-analysis result.

Six variables were investigated: baseplate length (100–200 mm), width (80–150 mm), thickness (4–15 mm), rib height (20–60 mm), rib thickness (3–10 mm), and fillet radius (2–10 mm). Four 10 mm bolt holes and two ribs were fixed features. Acceptance required $FoS \geq 2.5$ and $\delta \leq 0.5$ mm.

Mass was estimated from plate and rib volume less bolt holes. Stress and displacement were screened using equivalent cantilever relations:

$$M = F_d e, \quad \sigma_{max} = M/Z_{eff}, \quad FoS = \sigma_y / \sigma_{max}, \quad (3.2)$$

$$\delta = F_d e^3 / (3EI_{eff}). \quad (3.3)$$

These relations do not resolve bolt bearing, weld-toe stress, local contact, or three-dimensional restraint.

3.2 Padeye Definition

The padeye comprised an S275 steel doubler plate, tapered lug, 50 mm pin hole, twin gussets, and root fillets. A 500 kg working load and dynamic factor 3 produced 14,715 N pin-bearing load. Nine variables controlled the plate, lug, neck, gussets, and fillets. Acceptance required $FoS \geq 2.5$ and displacement below 0.5 mm.

3.3 Stabilizer Definition

The stabilizer retained a NACA 0015 envelope with 800 mm span, 500 mm root chord, 250 mm tip chord, and 12-degree sweep. Nine internal-structure variables controlled skins, spars, ribs, and root reinforcement. A 11,808 N resultant load was distributed over four spanwise pressure bands. Acceptance required $FoS \geq 2.5$ and tip displacement below 5 mm. Because pressure was prescribed rather than obtained from CFD, this was structural sizing rather than hydrodynamic optimization.

3.4 Sampling and Optimization

Thirty bracket designs were evaluated using Latin-hypercube sampling [20]. Two bounded Nelder–Mead searches of 80 iterations were then started from the baseline and best sampled design [10, 15]. Mass was penalized for normalized factor-of-safety or deflection violations. The result was interpreted as the best design found within the chosen variables and starts, not a guaranteed global optimum.

3.5 CAD and Validation Procedure

Fusion 360 generators rebuilt each component and exported STEP geometry and boundary metadata. Candidate packages identified fixed supports, loads, pressure bands, and monitoring points. Intended FEA verification used linear static studies with local mesh refinement at holes, fillets, roots, gussets, and spar intersections. The available checklists contain no completed Fusion stress, FOS, or displacement entries; analytical screening results are therefore reported separately from FEA validation.

Chapter 4

Results and Discussion

4.1 Bracket Results

The baseline used a 150 by 100 by 8 mm baseplate, 40 mm rib height, 6 mm rib thickness, and 5 mm fillet radius. Table 4.1 summarizes the analytical campaign.

Table 4.1: Analytical bracket results

Design	Mass (kg)	Stress (MPa)	FOS	Deflection (mm)
Baseline	0.4181	41.53	6.65	0.196
Best sample	0.2515	30.10	9.17	0.072
Baseline-start result	0.1265	108.11	2.55	0.476
Sample-start result	0.1126	106.30	2.60	0.361

The best sample reduced mass by 39.8% but remained lightly utilized. The baseline-start search required 132 evaluations and approached the deflection constraint. The sample-start search required 134 evaluations and produced the lightest screened design: 100.09 mm length, 80.00 mm width, 4.01 mm base thickness, 33.64 mm rib height, 3.00 mm rib thickness, and 4.11 mm fillet radius.

The predicted reduction was

$$\frac{0.418125 - 0.112593}{0.418125} \times 100 = 73.1\%. \quad (4.1)$$

Most dimensions approached their lower bounds. This reveals both the opportunity and risk of optimization: the algorithm exploited the simplified model and limits provided. Minimum plate gauge, bolt bearing, weldability, corrosion allowance, fatigue, vibration, and fabrication tolerance must be added before accepting such thin members.

4.2 Engineering Interpretation

Mass reduced with plate plan dimensions and thickness, while retained rib height preserved section depth and bending stiffness. The two searches converged to different rib-height and fillet combinations, confirming sensitivity to the starting point. Both final designs operated near the analytical FOS limit, so conservative backup candidates should accompany the lightest design during FEA.

4.3 Padeye and Stabilizer Outcomes

For the padeye, the project completed the load case, material model, nine-variable screening model, CAD generator, boundary definitions, and validation package. Critical regions are the pin-hole net section, bearing surface, lug root, gusset roots, and welded interfaces.

For the stabilizer, the project completed the fixed-envelope internal-structure model, pressure bands, analytical screening, CAD generator, and validation package. Critical regions are the root, spar connections, skin-spar intersections, and trailing edge. These extensions demonstrate method portability but not service-ready designs.

4.4 Validation Status

CAD generation and export were achieved, but recorded Fusion maximum stress, FOS, and displacement values were absent from the validation checklists. All numerical structural results in Table 4.1 are therefore analytical predictions. No claim of classification approval, FEA compliance, or fabrication readiness is made.

Chapter 5

Conclusions and Recommendations

The project established a traceable engineering workflow for parameterized marine components. Bracket sampling and optimization reduced analytical mass from 0.4181 kg to 0.1126 kg while retaining predicted FOS 2.60 and deflection 0.361 mm. Engineering definitions and validation workflows were also completed for a steel lifting padeye and aluminium stabilizer.

The 73.1% reduction is a screening result, not a validated design saving. The analytical model omits local and cyclic failure modes, and Fusion structural results were not recorded. The lightest bracket must therefore not be fabricated or installed on the basis of this report alone.

5.1 Recommendations

1. Complete mesh-converged FEA for every baseline and at least three finalists.
2. Add bolt bearing, tear-out, weld-group, local plate, fatigue, buckling, and corrosion checks.
3. Proof-load fabricated bracket and padeye specimens and measure strain and displacement.
4. Derive stabilizer pressure from CFD, model testing, or an accepted hydrodynamic method.
5. Apply relevant classification-society and statutory requirements before vessel use.

5.2 Future Work

Padeye development should include pin contact, net-section failure, weld fatigue, and proof loading. Stabilizer development should combine hydrodynamic pressure, structural

FEA, buckling, fatigue, and root-bearing assessment. Propeller work should begin only after a defensible thrust, torque, efficiency, cavitation, and pressure model is available.

Appendix A

Appendices

A.1 Required Validation Record

Each final FEA record should include material assignment, mesh sizes and element count, boundary-condition screenshots, reaction-force balance, maximum von Mises stress, minimum FOS, maximum displacement, convergence evidence, mass, pass/fail status, and engineering comments.

A.2 Reproducibility

The reported bracket campaign comprised 30 sampled evaluations, 132 evaluations in the baseline-start search, and 134 evaluations in the sample-start search. Source CSV and JSON records are retained in the formal run directories.

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